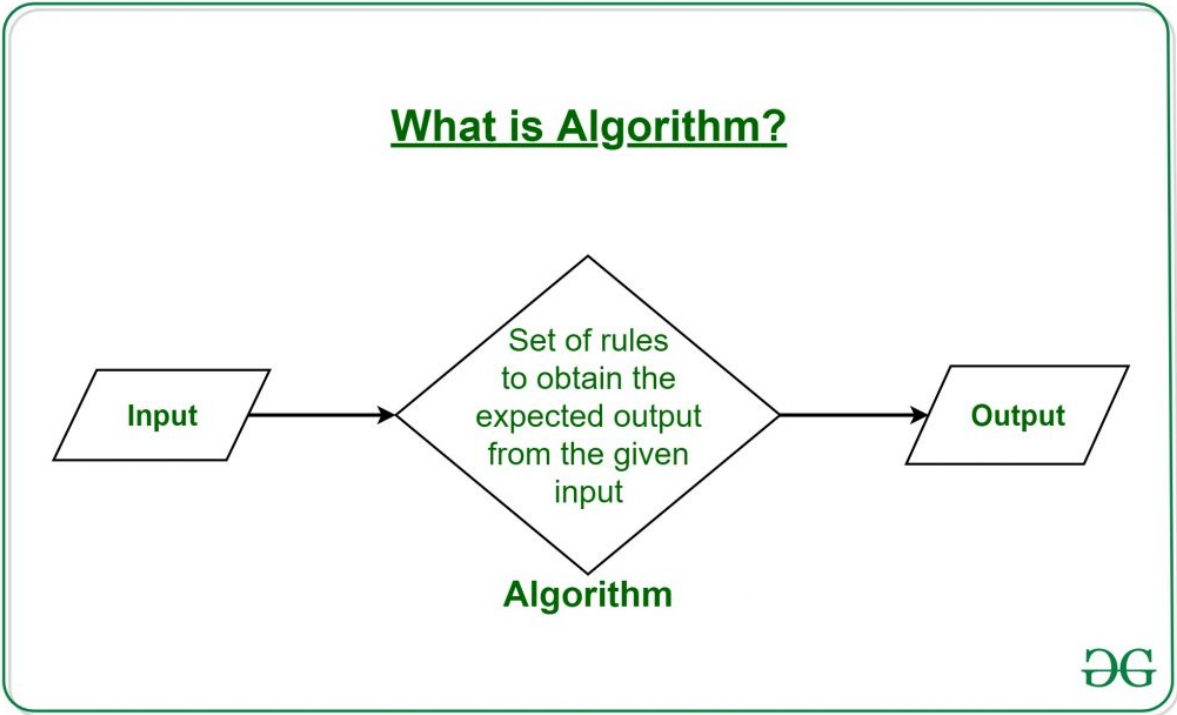


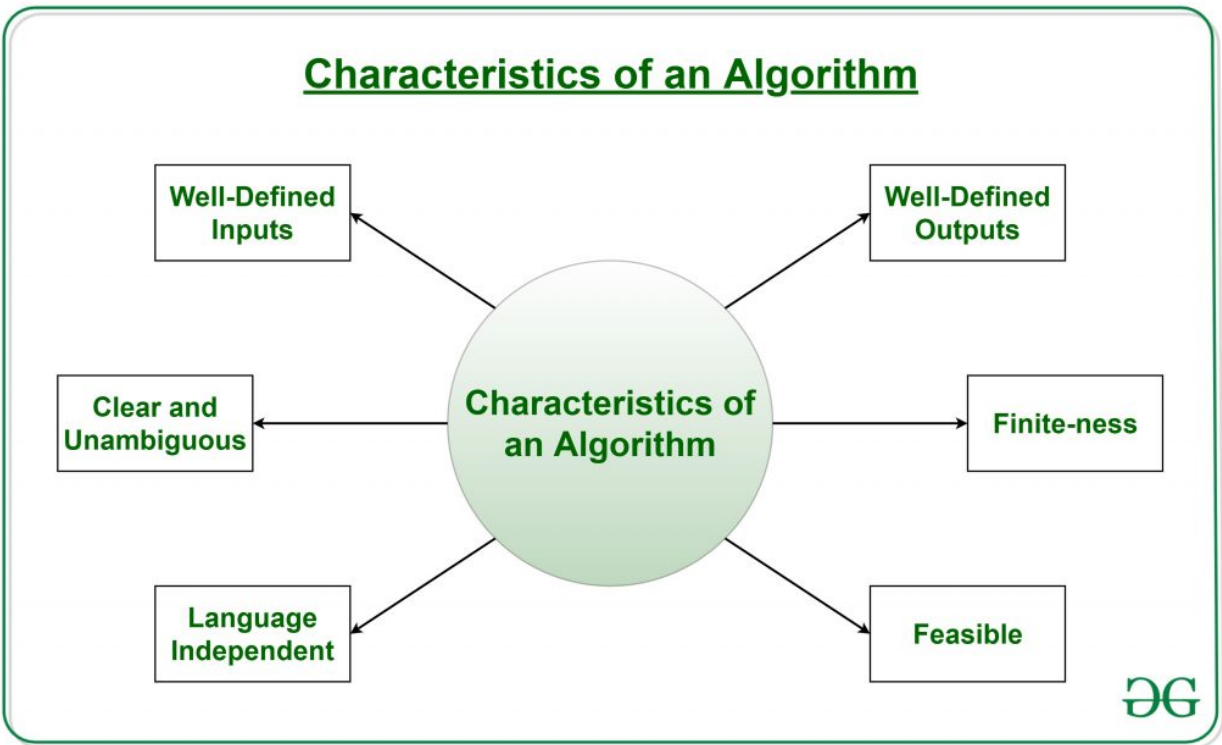
INTRODUCTION TO ALGORITHMS

The word Algorithm means “a process or set of rules to be followed in calculations or other problem-solving operations”. Therefore Algorithm refers to a set of rules/instructions that step-by-step define how a work is to be executed upon in order to get the expected results.



It can be understood by taking an example of cooking a new recipe. To cook a new recipe, one reads the instructions and steps and execute them one by one, in the given sequence. The result thus obtained is the new dish cooked perfectly. Similarly, algorithms help to do a task in programming to get the expected output.

The Algorithm designed are language-independent, i.e. they are just plain instructions that can be implemented in any language, and yet the output will be the same, as expected.



Characteristics:

- Clear and Unambiguous: Algorithm should be clear and unambiguous.
- Well-Defined Inputs: If an algorithm says to take inputs, it should be well-defined inputs.
- Well-Defined Outputs: The algorithm must clearly define what output will be yielded
- Finite-ness: The algorithm must be finite, i.e. it should not end up in an infinite loops or similar.
- Feasible: The algorithm must be simple, generic and practical, such that it can be executed upon will the available resources. It must not contain some future technology, or anything.
- Language Independent: The Algorithm designed must be language-independent, i.e. it must be just plain instructions.

How to Design an Algorithm?

Inorder to write an algorithm, following things are needed as a pre-requisite:

- The problem that is to be solved by this algorithm.
- The constraints of the problem that must be considered while solving the problem.
- The input to be taken to solve the problem.
- The output to be expected when the problem the is solved.
- The solution to this problem, in the given constraints.

Then the algorithm is written with the help of above parameters such that it solves the problem.

PROBLEM-1 : Algorithm to add 3 numbers and print their sum:

1. START
2. Declare 3 integer variables num1, num2 and num3.
3. Take the three numbers, to be added, as inputs in variables num1, num2, and num3 respectively.
4. Declare an integer variable sum to store the resultant sum of the 3 numbers.
5. Add the 3 numbers and store the result in the variable sum.
6. Print the value of variable sum
7. STOP

PROBLEM-2 : Algorithm to input any 2 numbers and print their sum and multiplication.

1. START
2. Declare any 2 integer variables A and B.
3. Take the two numbers, to be added and multiplied, as inputs in variables A and B respectively.
4. Declare two integer variables SUM and MULT store the result.
5. Add and Multiply the 2 numbers and store the result into SUM and MULT respectively
6. Output/ Print the values of SUM and MULT
7. STOP

QUESTIONS :

1. Define algorithm ? Whar are its characteristics ?
2. Write an algorithm to input the integer values for 3 variables and print their SUM and AVERAGE.

(To be sent through WhatsApp Group)
